

Problem 1

a)

We consider the state with basis $|0\rangle, |1\rangle, \dots, |d-1\rangle$, and have another orthonormal basis $|v_0\rangle, |v_1\rangle, \dots, |v_{d-1}\rangle$. To show that there is a unitary transformation U such that $U|i\rangle = |v_i\rangle$ we start with an operator $|v_i\rangle\langle i|$ and check how it acts on a vector $|j\rangle$.

$$(|v_i\rangle\langle i|)|j\rangle = |v_i\rangle\langle i|j\rangle = \langle i|j\rangle |v_i\rangle = \delta_{ij} |v_i\rangle \quad (1)$$

This gives that the operator $|v_i\rangle\langle i|$ will return the vector $|v_i\rangle$ when applied to $|i\rangle$ and 0 otherwise. Taking advantage of this attribute, we construct the transformation by taking

$$U = \sum_{i=0}^{d-1} |v_i\rangle\langle i| \quad (2)$$

To show that U is a unitary operator we check

$$UU^\dagger = \sum_{i=0}^{d-1} |v_i\rangle\langle i| \sum_{j=0}^{d-1} |j\rangle\langle v_j| = \sum_{i,j=0}^{d-1} |v_i\rangle\langle i|j\rangle\langle v_j| = \sum_{i,j=0}^{d-1} |v_i\rangle\delta_{ij}\langle v_j| = \sum_{i=0}^{d-1} |v_i\rangle\langle v_i| = I$$

b)

We see what the operator does to an arbitrary vector of the basis: $\sum_{i=0}^{d-1} c_i |v_i\rangle$

$$\sum_{i=0}^{d-1} |v_i\rangle\langle v_i| \sum_{j=0}^{d-1} c_j |v_j\rangle = \sum_{i=0}^{d-1} \sum_{j=0}^{d-1} c_j |v_i\rangle\langle v_i|v_j\rangle = \sum_{i=0}^{d-1} \sum_{j=0}^{d-1} \delta_{ij} c_j |v_i\rangle = \sum_{i=0}^{d-1} c_i |v_i\rangle\langle v_i| \quad (3)$$

i.e

$$\sum_{i=0}^{d-1} |v_i\rangle\langle v_i| = I \quad (4)$$

Problem 2

a)

Show that the matrix U is unitary, i.e. $U^\dagger U = I$

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \quad (5)$$

$$U^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \quad (6)$$

$$U^\dagger U = UU^\dagger = \frac{1}{2} \begin{pmatrix} 1*1 + i*(-i) & 1*1 + i*i \\ 1*1 + (-i)*(-i) & 1*1 + i*(-i) \end{pmatrix} = I \quad (7)$$

b)

How U acts upon the states: $|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+\rangle$ and $|-\rangle$ we get:

$$U|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle \quad (8)$$

$$U|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ -i \end{pmatrix} = i|-\rangle \sim |-\rangle \quad (9)$$

$$U|+\rangle = \frac{e^{-\frac{i\pi}{4}}}{2} \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} = e^{-\frac{i\pi}{4}} |+\rangle \quad (10)$$

$$U|-\rangle = \frac{e^{\frac{i\pi}{4}}}{2} \begin{pmatrix} 1-i \\ 1+i \end{pmatrix} = e^{\frac{i\pi}{4}} |-\rangle \quad (11)$$

$$U|+\rangle = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \quad (12)$$

$$U|-\rangle = \frac{1}{2} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \quad (13)$$

We see that the unitary operator U transforms $|0\rangle \rightarrow |+\rangle, |+\rangle \rightarrow |i\rangle, |i\rangle \rightarrow |0\rangle$, i.e. $z \rightarrow x, x \rightarrow y, y \rightarrow z$

c)

A pure qubit has the density matrix $\rho = \frac{1}{2}(I + r \cdot \sigma)$, with $r = (x, y, z)$ being its Bloch vector. From our results in the previous task, we see that U rotates the Bloch sphere, $(x, y, z) \rightarrow (z, x, y)$. From this, we easily construct the matrix:

$$R = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad (14)$$

We solve the equation $Rn = n$ to find the axis of rotation.

$$Rn = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix} = \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix} \quad (15)$$

giving the solution $n = \frac{1}{\sqrt{3}}(1, 1, 1)$, i.e. axis $(1, 1, 1)$

Problem 3

Let $|0\rangle, |1\rangle, |2\rangle$ be a basis for the system.

a)

Firstly we need to check the unit length of the basis. Using the characteristic $\langle x_i | x_j \rangle = \delta_{ij}, \forall x_{i,j} \in |0\rangle, |1\rangle, |2\rangle$ we get the following inner products:

$$\langle a | a \rangle = \frac{1}{2} \langle 0 | 0 \rangle + \frac{1}{4} \langle 1 | 1 \rangle + \frac{1}{4} \langle 2 | 2 \rangle = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \quad (16)$$

$$\langle b | b \rangle = \frac{1}{2} \langle 0 | 0 \rangle + \frac{1}{4} \langle 1 | 1 \rangle + \frac{1}{4} \langle 2 | 2 \rangle = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \quad (17)$$

$$\langle c | c \rangle = \frac{1}{2} \langle 1 | 1 \rangle + \frac{1}{2} \langle 2 | 2 \rangle = \frac{1}{2} + \frac{1}{2} = 1 \quad (18)$$

There can only exist three(3) orthonormal vectors in a three dimensional space, so we test inner products.

$$\langle a | b \rangle = \frac{1}{2} \langle 0 | 0 \rangle - \frac{1}{4} \langle 1 | 1 \rangle - \frac{1}{4} \langle 2 | 2 \rangle = \frac{1}{2} - \frac{1}{4} - \frac{1}{4} = 0 \quad (19)$$

$$\langle b | c \rangle = -\frac{\sqrt{2}-1}{2} \langle 1 | 1 \rangle + \frac{\sqrt{2}-1}{2} \langle 2 | 2 \rangle = -\frac{\sqrt{2}-1}{2} + \frac{\sqrt{2}-1}{2} = 0 \quad (20)$$

$$\langle a | c \rangle = \frac{\sqrt{2}-1}{2} \langle 1 | 1 \rangle - \frac{\sqrt{2}-1}{2} \langle 2 | 2 \rangle = \frac{\sqrt{2}-1}{2} - \frac{\sqrt{2}-1}{2} = 0 \quad (21)$$

i.e. $|a\rangle, |b\rangle, |c\rangle$ is an orthonormal basis.

b)

The von neumann method of measurement is using a self-adjoint operator named an observable to conduct the measurement. We take the projection of the state onto the basis:

$$\langle \psi | a \rangle = \frac{\sqrt{2}}{\sqrt{6}} \langle 0 | 0 \rangle + \frac{\sqrt{2}}{\sqrt{6}} \langle 1 | 1 \rangle = \sqrt{\frac{2}{3}} \quad (22)$$

$$\langle \psi | b \rangle = \frac{\sqrt{2}}{\sqrt{6}} \langle 0 | 0 \rangle - \frac{\sqrt{2}}{\sqrt{6}} \langle 1 | 1 \rangle = 0 \quad (23)$$

$$\langle \psi | c \rangle = \frac{\sqrt{2}}{\sqrt{6}} \langle 1 | 1 \rangle = \sqrt{\frac{1}{3}} \quad (24)$$

I.e. we get probability $\frac{2}{3}$ of being in $|a\rangle$, 0 for $|b\rangle$ and $\frac{1}{3}$ of being in state $|c\rangle$.

For measurement of $|a\rangle, |b\rangle$ and $|c\rangle$ the state collapses to subsequently $|a\rangle, |b\rangle$ and $|c\rangle$

Problem 4

a)

Probability of transmission through a polarizer is given by Malus' law: $P = \cos^2(\Delta\theta)$ If we assume completely unpolarised light before the first filter, photons must rotate by 45 degrees three(3) times. the amount of light passing through will be $\frac{1}{2} \cdot \cos^2(45^\circ) \cdot \cos^2(45^\circ) = \frac{1}{8}$

b)

For the next setup, we have two options: [1] : $0^\circ \rightarrow 30^\circ \rightarrow 60^\circ \rightarrow 90^\circ$ or [2] : $0^\circ \rightarrow 60^\circ \rightarrow 30^\circ \rightarrow 90^\circ$.

First we check option [1]:

$$P = \frac{1}{2} \cdot \cos^2(30^\circ) \cdot \cos^2(30^\circ) \cdot \cos^2(30^\circ) \approx 0.211$$

and option [2]:

$$P = \frac{1}{2} \cdot \cos^2(60^\circ) \cdot \cos^2(30^\circ) \cdot \cos^2(60^\circ) \approx 0.0234$$

So option [1] will let substantially more light through.

Problem 5

a)

We construct each object by combining a polarizing filter P with a unitary transformation U . We want the unitary to have the property that light passing through the first polarizer and then through U has probability $1/2$ of passing through any linear polarizer, independent of its orientation.

We choose our basis such that the first polarizer transmits

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Suppose that

$$U|0\rangle = \begin{pmatrix} a \\ b \end{pmatrix}.$$

A linear polarizer rotated by an arbitrary angle θ transmits the state

$$|\theta\rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}.$$

We require the probability of passing through this polarizer to be $1/2$ for every θ , such that

$$|\langle \theta | U | 0 \rangle|^2 = |a \cos \theta + b \sin \theta|^2 = \frac{1}{2}.$$

Expanding this expression gives

$$|a|^2 \cos^2 \theta + |b|^2 \sin^2 \theta + 2 \operatorname{Re}(a^* b) \sin \theta \cos \theta = \frac{1}{2}.$$

For this to hold independently of θ , we require

$$|a|^2 = |b|^2 = \frac{1}{2}, \quad \operatorname{Re}(a^* b) = 0.$$

A simple choice satisfying these conditions is

$$a = \frac{1}{\sqrt{2}}, \quad b = \frac{i}{\sqrt{2}},$$

and hence

$$U|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

This determines the first column of U . Since U must be unitary, its second column must be a normalized vector orthogonal to the first. One possible choice is

$$\frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix},$$

giving

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Indeed, $U^\dagger U = I$.

We now construct each object from one such unitary transformation and one polarizing filter. In the $ABCD$ configuration, the total transformation can be written as

$$UP_1P_2U^\dagger.$$

The unitary transformations do not change the intensity, so the transmission probability is the same as for P_1P_2 . Thus, the two objects behave like ordinary polarizing filters in this orientation.

If one object is flipped, the essential part of the transformation becomes

$$P_1UP_2.$$

After passing through the first polarizer, the photon is in the state $|0\rangle$ in the coordinate system of that object. Applying U gives

$$U|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

The probability of this state passing through the second polarizer at an arbitrary relative angle θ is

$$\begin{aligned} |\langle\theta|U|0\rangle|^2 &= \frac{1}{2} |\cos\theta + i\sin\theta|^2 \\ &= \frac{1}{2} (\cos^2\theta + \sin^2\theta) \\ &= \frac{1}{2}. \end{aligned}$$

Thus, this probability is independent of the relative angle θ . Since the first polarizer transmits $1/2$ of the initially unpolarized light, the total fraction transmitted is

$$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Therefore, this construction produces the required 25% transmission independent of the relative orientation of the two objects.

b)

If we order the objects as $BADC$, we would get the effect of $PUU^\dagger P$. Using the fact that U is unitary, we just get PP , i.e. the effect of just two normal filters next to each other